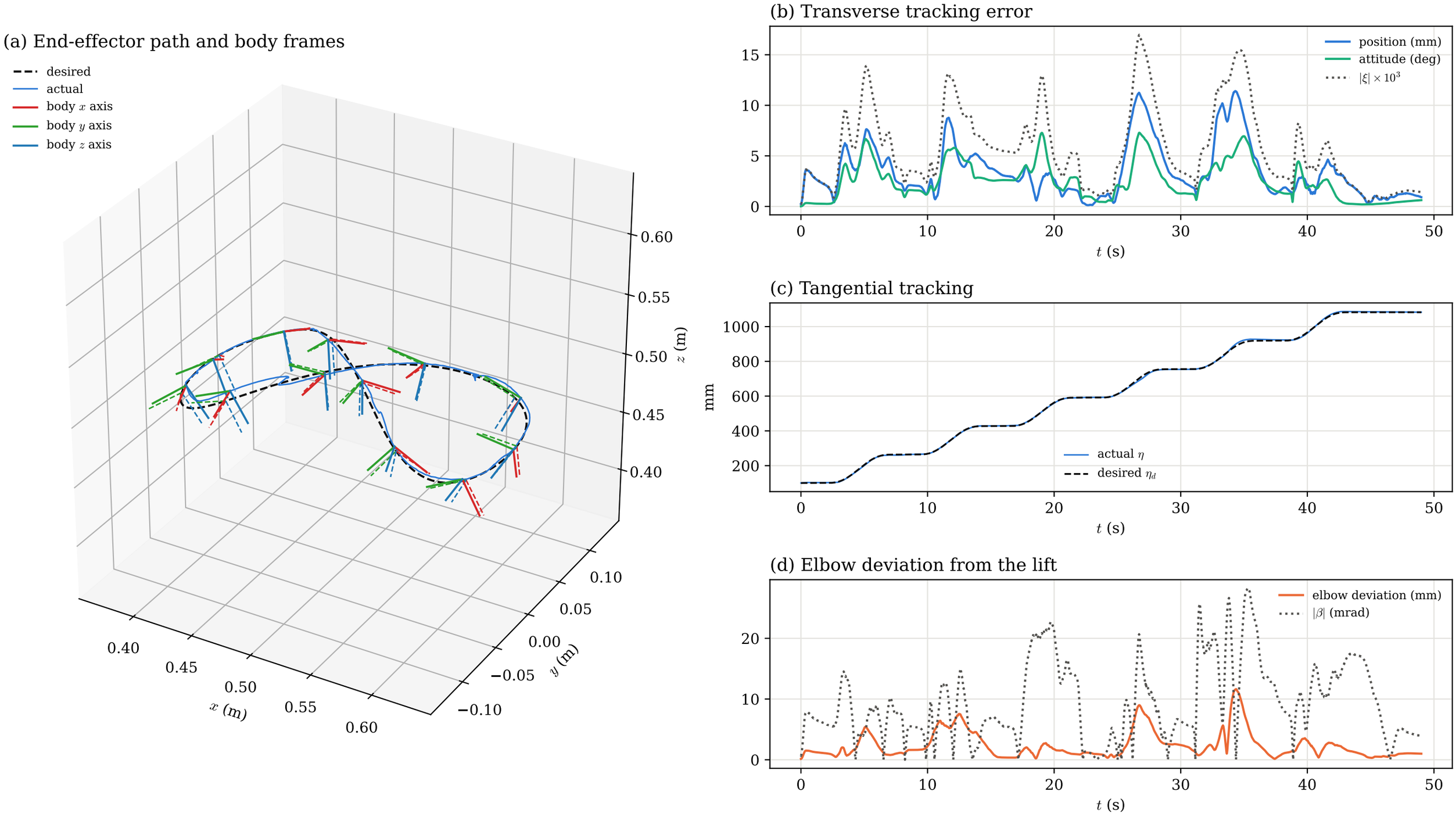

(a) End-effector path and body frames
desired
actual
body x axis
body y axis
body z axis
0.40
0.45
0.50
0.55
0.60
x (m)
0.10
0.05
0.00
−0.05
−0.10
y (m)
0.60
0.55
0.50
0.45
0.40
z (m)
(b) Transverse tracking error
position (mm)
attitude (deg)
$|\xi| \times 10^3$
0
5
10
15
0
10
20
30
40
50
t (s)
(c) Tangential tracking
mm
200
400
600
800
1000
actual η
desired η_d
0
10
20
30
40
50
t (s)
(d) Elbow deviation from the lift
elbow deviation (mm)
$|\beta|$ (mrad)
0
10
20
0
10
20
30
40
50
t (s)